SVM Tutorial: Min–Max, the Dual and the Kernel Trick

Name | | | | | | | | | | | | | | | | | | | | | | |

Roll No. | | | | | | | | | *one digit per box*

Work individually. Ten questions, each building on the last. Write your working in the boxes.

Part A. Who moves first?

A parent chooses the *shop* and wants a low price; a kid chooses the *toy* and wants a high price.

Q1. Prices (Rs.):

	Toy 1	Toy 2
Shop A	50	80
Shop B	70	60

(a) Parent first: compute $\min_s \max_t p(s,t)$. (b) Kid first: compute $\max_t \min_s p(s,t)$. (c) Which is smaller, and why?

Q2. Prices (Rs.):

	Toy 1	Toy 2	Toy 3
Shop A	3	5	4
Shop B	6	7	8
Shop C	2	9	1

Compute min–max and max–min. Find the entry that is the *largest in its row* and the *smallest in its column.*

Facts we will use

- **Weak duality:** for any table, $\max\min \le \min\max$.
- **Strong duality:** at a saddle point, $\max\min = \min\max$. The SVM problem always has one.

Part B. From the story to the SVM

For data $(\mathbf{x}_i, y_i)$, $y_i \in \{-1, +1\}$, the SVM "bill" is $L(\mathbf{w}, b, \alpha) = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_i \alpha_i \big[y_i(\mathbf{w}\cdot\mathbf{x}_i + b) - 1\big]$, with $\alpha_i \ge 0$.

Q3. Complete the table.

	Variables	Wants L	Name / problem
Parent			
Kid			
Primal			
Dual			

Q4. Take $\mathbf{w} = (1, 1)$, $b = -3$. Compute $g_i = y_i(\mathbf{w} \cdot \mathbf{x}_i + b) - 1$ and say what the kid does with α_i.

i	$\mathbf{x}_i$	y_i	$\mathbf{w} \cdot \mathbf{x}_i + b$	g_i	Kid's action
1	$(3, 2)$	$+1$			
2	$(2, 2)$	$+1$			
3	$(2, 0)$	$+1$			
4	$(0, 1)$	-1			

Facts we will use

Solving the parent's reply and substituting back gives the **dual problem** and the **classifier**:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j \, y_i y_j \, K(\mathbf{x}_i, \mathbf{x}_j) \quad \text{s.t. } \alpha_i \geq 0, \;\; \sum_i \alpha_i y_i = 0; \qquad f(\mathbf{x}) = \sum_i \alpha_i y_i \, K(\mathbf{x}_i, \mathbf{x}) + b.$$

With $K(\mathbf{x}, \mathbf{z}) = \mathbf{x} \cdot \mathbf{z}$ this is the ordinary (linear) SVM, and then $\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$. The **kernel trick**: replace K by any other kernel and nothing else changes. Support vectors satisfy $y_i f(\mathbf{x}_i) = 1$. In the **Gaussian (RBF) kernel** $K(\mathbf{x}, \mathbf{z}) = e^{-\gamma \|\mathbf{x} - \mathbf{z}\|^2}$, the constant $\gamma > 0$ is a **hyperparameter**: chosen by the user, not learned. It sets how quickly similarity decays with distance (similarity drops to $e^{-1} \approx 0.37$ at distance $1/\sqrt{\gamma}$).

Part C. The kernel in practice

Definition (Gram matrix). Given points $\mathbf{x}_1, \ldots, \mathbf{x}_n$ and a kernel K, the **Gram matrix** is the $n \times n$ table of all pairwise kernel values,

$$\mathbf{K} = \begin{pmatrix} K(\mathbf{x}_1, \mathbf{x}_1) & \cdots & K(\mathbf{x}_1, \mathbf{x}_n) \\ \vdots & \ddots & \vdots \\ K(\mathbf{x}_n, \mathbf{x}_1) & \cdots & K(\mathbf{x}_n, \mathbf{x}_n) \end{pmatrix}, \qquad K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j).$$

Entry K_{ij} measures how similar points i and j are. It is all the dual problem needs from the data: training an SVM means working with this one matrix.

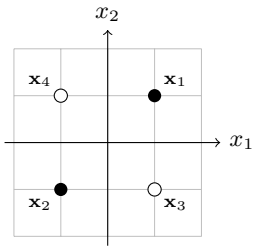
Q5. A complete SVM by hand (linear kernel). Points $\mathbf{x}_1 = (1, 1)$, $y_1 = +1$ and $\mathbf{x}_2 = (-1, -1)$, $y_2 = -1$. (a) Use $\sum_i \alpha_i y_i = 0$ to show $\alpha_1 = \alpha_2 = \alpha$. (b) Write the dual objective as a function of α and maximise it. (c) Find $\mathbf{w}$ and b, and write $f(\mathbf{x})$.

Q6. Gram matrices. Using the definition above, for $\mathbf{x}_1 = (1, 0)$, $\mathbf{x}_2 = (0, 1)$, $\mathbf{x}_3 = (1, 1)$, write the 3×3 matrix $K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$ for (a) $K(\mathbf{x}, \mathbf{z}) = \mathbf{x} \cdot \mathbf{z}$ and (b) $K(\mathbf{x}, \mathbf{z}) = (\mathbf{x} \cdot \mathbf{z})^2$. What is K_{23} in each case?

Q7. The trick itself. For $K(\mathbf{x}, \mathbf{z}) = (\mathbf{x} \cdot \mathbf{z})^2$ the hidden features are $\phi(\mathbf{x}) = (x_1^2,\ \sqrt{2}\, x_1 x_2,\ x_2^2)$. For $\mathbf{x} = (2, 1)$, $\mathbf{z} = (1, 3)$ compute $K(\mathbf{x}, \mathbf{z})$ directly, then compute $\phi(\mathbf{x}) \cdot \phi(\mathbf{z})$. What do you notice?

Q8. Kernel trick in action: XOR. $\mathbf{x}_1 = (1, 1)$, $\mathbf{x}_2 = (-1, -1)$ have $y = +1$ (filled); $\mathbf{x}_3 = (1, -1)$, $\mathbf{x}_4 = (-1, 1)$ have $y = -1$ (hollow). Use $K(\mathbf{x}, \mathbf{z}) = (\mathbf{x} \cdot \mathbf{z})^2$.

(a) Look at the figure. Can any straight line separate the two classes?
(b) Compute the 4×4 Gram matrix.
(c) Take $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = a$. Check $\sum \alpha_i y_i = 0$, write the dual objective in terms of a, and maximise.
(d) Using $\mathbf{x}_1$, find b. Then write $f(\mathbf{x}) = \sum_i \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$ and simplify it in terms of x_1, x_2.
(e) Classify $(2, 3)$ and $(-1, 2)$ using *only kernel evaluations*, and check against (d).



(a)

(b)

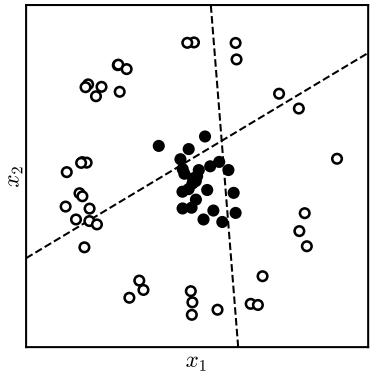
(c)

(d)

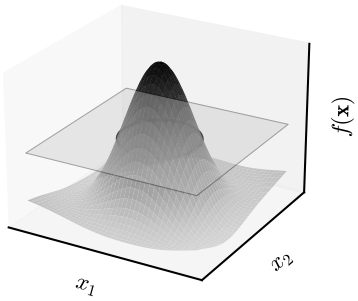
(e)

Illustration: the Gaussian kernel before and after.

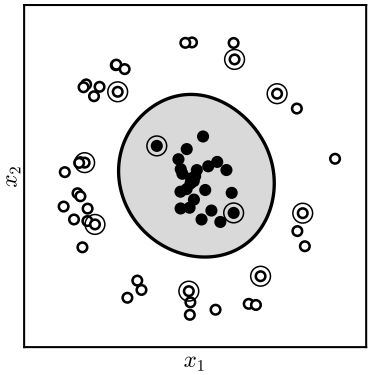
(a) Before: no straight line works



(b) Kernel trick: $f(\mathbf{x})$ as a landscape, flat plane $f=0$ is sea level



(c) After: boundary $f(\mathbf{x})=0$



(a) *Data.* 25 points with $y_i = +1$ (filled) satisfy $\|\mathbf{x}_i\| \le 0.8$; 40 points with $y_i = -1$ (hollow) satisfy $1.6 \le \|\mathbf{x}_i\| \le 2.1$. No $(\mathbf{w}, b)$ achieves $y_i(\mathbf{w}\cdot\mathbf{x}_i + b) > 0$ for all i, so a linear SVM fails.

(b) *Kernel.* With $K(\mathbf{x},\mathbf{z}) = e^{-0.5\|\mathbf{x}-\mathbf{z}\|^2}$ the trained SVM is $f(\mathbf{x}) = \sum_{i\in\mathrm{SV}} \alpha_i y_i\, e^{-0.5\|\mathbf{x}_i-\mathbf{x}\|^2} + b$. Each term with $y_i = +1$ is a Gaussian bump centred at $\mathbf{x}_i$; each term with $y_i = -1$ is a Gaussian dip. The surface shows $f(\mathbf{x})$ as height; the flat plane is $f = 0$.

(c) *Result.* The classifier is $\hat{y} = \operatorname{sign} f(\mathbf{x})$, and the boundary $\{\mathbf{x} : f(\mathbf{x}) = 0\}$ is a closed curve around the $+1$ points. Only 10 of the 65 points have $\alpha_i > 0$ (circled). The feature map ϕ is never formed: f needs only the values $K(\mathbf{x}_i, \mathbf{x})$.

Q9. RBF kernel as similarity. For $K(\mathbf{x},\mathbf{z}) = e^{-\gamma\|\mathbf{x}-\mathbf{z}\|^2}$, $\mathbf{x} = (0,0)$, $\mathbf{z} = (1,1)$, compute K for $\gamma = 0.5$ and $\gamma = 5$. (Use $e^{-1} \approx 0.37$, $e^{-10} \approx 0.00005$.) Which γ says "these points are similar"?

Q10. Predicting with an RBF SVM. A trained SVM ($\gamma = 1$, $b = 0$) has two support vectors: $\mathbf{x}_1 = (0,0)$, $y_1 = +1$, $\alpha_1 = 1$ and $\mathbf{x}_2 = (2,0)$, $y_2 = -1$, $\alpha_2 = 1$. So $f(\mathbf{x}) = K(\mathbf{x}_1,\mathbf{x}) - K(\mathbf{x}_2,\mathbf{x})$. (a) Classify $(0.5,0)$ and $(1.5,0)$. (Use $e^{-0.25} \approx 0.78$, $e^{-2.25} \approx 0.11$.) (b) What is $f(1,0)$? (c) Describe the decision boundary.